



Major Project

Title: Kolmogorov-Arnold Networks in Recurrent Recommenders: Session-Based and Next-Basket Prediction

End-Term Evaluation

Presented by:

Abhishek Yadav | R2142231410 | SAP 500124297

Saksham Agrawal | R2142231383 | SAP 500121969

Yash Dagar | R2142231424 | SAP 500125147

Mentored by:

Dr. Amar Jindal

Assistant Professor,

Data Science Cluster

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1. Introduction: Technical Background

Session-based recommendation

Anonymous browsing session

- Input: the clicks so far, $i_1, i_2, \dots, i_t$
- Output: a ranking of all items for the next click
- No user profile: visitors are anonymous
- Model: GRU4Rec (gated recurrent unit)
- Data: YooChoose (RecSys Challenge 2015)

Next-basket recommendation

Returning grocery customer

- Input: the customer's past baskets $B_1 \dots B_T$
- Output: a ranking of all items for basket B_{T+1}
- Long histories, most items are re-purchases
- Model: basket-GRU (van Maasakkers et al., 2023)
- Data: Instacart, Dunnhumby Complete Journey

Common core: both tasks use a GRU. This project asks where, if anywhere, Kolmogorov-Arnold Network (KAN) layers improve these GRU recommenders when compared fairly.

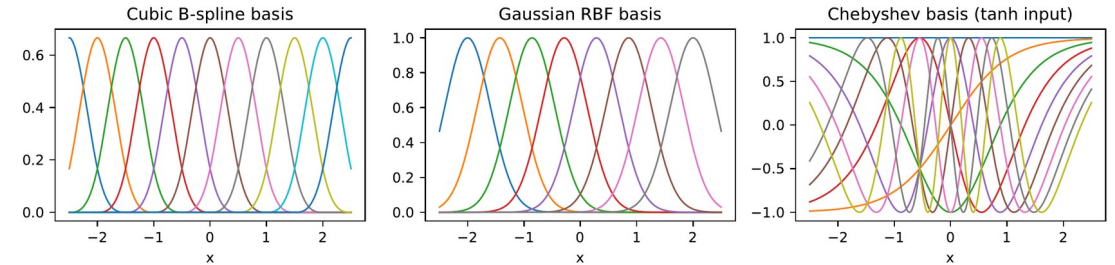
1. Introduction: Technical Concepts Used

Kolmogorov-Arnold Network (KAN) layer

A learnable univariate function on every edge:

$$y_o = \sum_i \varphi_{o,i}(x_i), \quad \varphi(x) = w \cdot \text{silu}(x) + \sum_k c_k B_k(x)$$

- Four basis families B_k implemented: cubic B-spline, Gaussian RBF (FastKAN), Chebyshev, group-rational (GR-KAN)
- GRU4Rec: official PyTorch code, session-parallel mini-batches, 2048 shared negatives, cross-entropy with logQ correction
- Basket-GRU: multi-hot basket in, sigmoid score for every product out
- Parameter-matched MLP control for every KAN block, so gains from extra parameters are not credited to the KAN



Basis functions used inside the KAN layers (from our code)

Residual KAN block

$$h' = h + \text{KAN}(\text{LayerNorm}(h))$$

Output layer initialised to zero, so each variant starts exactly as its baseline.

1. Introduction: Motivation and Problem Statement

Motivation

- Anonymous and privacy-restricted sessions make profile-based recommenders unusable
- KANs are claimed to be more accurate per parameter and more interpretable than MLPs
- Later studies (Yu et al. 2024; Hou et al. 2025) find KANs rarely win when parameters are matched
- Our first draft compared KAN and GRU on a leaky split with a broken B-spline, so its numbers could not support any conclusion

Problem Statement

Determine whether, and where, KAN layers improve GRU-based session and next-basket recommenders, under a protocol that allows a fair conclusion:

- standard, leakage-free splits comparable with published work
- verified implementations of baselines and KAN layers
- parameter-matched MLP controls for every KAN
- evaluation on both session and basket data

1. Introduction: Applications and Datasets

Area of application

- 1

E-commerce
 next product in a browsing session
- 2

Online grocery
 pre-filled next basket, reminders
- 3

Streaming and news
 next video, song or article
- 4

Privacy-first platforms
 no long-term tracking needed

Datasets and input format

	YooChoose 1/64	Instacart	Dunnhumby
Task	next click	next basket	next basket
Raw input	session, time, item	user, order, product	household, basket, day, product
Model input	item ID sequence	multi-hot baskets	multi-hot baskets
Users / sessions	139,628 sessions	20,620 users*	2,478 households
Items	17,264	12,058 groups	10,370
Test targets	55,357 clicks	10,310 baskets	1,239 baskets
Split	by time (last day)	last basket, users 50/50	last basket, users 50/50

* Mac profile: random 10% of Instacart users; the GPU profile uses all 206,209. Preprocessing follows GRU4Rec / SR-GNN and van Maasakkers et al. (2023).

2. Literature Review: Related Work

Work	Year	Model	Relevance / limitation
Hidasi et al.	2016/18	GRU4Rec	Strong session baseline; tuned version used here
Hidasi & Czapp	2023	Reimplementation study	Third-party GRU4Rec ports score 30-99% lower
Wu et al. (SR-GNN)	2019	Graph NN	Standard yoochoose1/64 benchmark table
Hu et al. (TIFU-KNN)	2020	Frequency + kNN	Simple model beats most neural NBR models
Li et al.	2023	NBR reality check	Repeat items dominate next-basket accuracy
van Maasakkers et al.	2023	Basket-GRU	Our basket baseline and protocol
Liu et al. (KAN)	2025	KAN	Learnable edge functions; slow to train
Yu et al.	2024	KAN vs MLP	With matched size, KAN wins only on symbolic formulas
Genet & Inzirillo (TKAN)	2024	KAN in LSTM	KAN inside a gated recurrent cell
Park et al. (CF-KAN)	2024	KAN autoencoder	Only KAN recommender with a matched MLP
Xu et al. (KANRec)	2026	KAN interest fusion	KAN in sequential recommendation

2. Literature Review: Inference and Research Gap

Session-based models

- GRU4Rec remains a strong baseline when the official code is used
- Weak GRU4Rec results in many papers come from flawed reimplementations

Next-basket models

- Repeat purchases carry most of the signal
- Personal-frequency baselines and TIFU-KNN are very hard to beat

KANs

- Interpretable learned functions, but slower than MLPs
- Gains often vanish against an MLP of the same size

Research gap

- No published work places KAN layers inside GRU4Rec or applies KANs to next-basket recommendation
- Few KAN recommenders control for model size with a parameter-matched MLP

Our question: where in a GRU recommender does a KAN layer help, compared with an MLP of the same size, on standard benchmarks?

2. Literature Review: SWOT Analysis

S Strengths

- Official GRU4Rec and published protocols
- Parameter-matched MLP control for every KAN
- Unit-tested KAN layer with four bases
- Three datasets, two tasks

W Weaknesses

- KAN layers are slower to train
- Extra hyperparameters (basis, grid size)
- Mac runs limited to one seed and a 10% Instacart sample

O Opportunities

- Interpretable models for repeat purchasing (KAN-NBR)
- GPU runs on the UPES Blackwell cluster
- KAN in attention models (SASRec)

T Threats

- Results may show no KAN advantage over MLPs
- Transformer recommenders dominate recent work
- Compute time for many variants

3. Objectives

Main objective

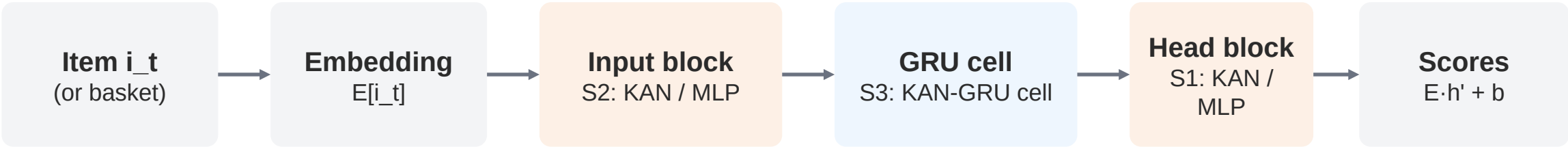
Build and fairly evaluate KAN-augmented GRU recommenders for session-based and next-basket prediction, and report where KAN layers help and where they do not.

Sub-objectives

- 1 Correct the methodological and reporting errors of the first draft
- 2 Implement and unit-test a KAN layer with B-spline, RBF, Chebyshev and rational bases
- 3 Add KAN layers to official GRU4Rec in three places: head, input, GRU cell
- 4 Apply the same placements to the basket-GRU on Instacart and Dunnhumby
- 5 Compare with parameter-matched MLPs and strong non-neural baselines, with significance tests
- 6 Propose KAN-NBR, an interpretable additive KAN over repeat-purchase features

4. Methodology: Reference Software Model

KAN placements in GRU4Rec (same placements in the basket-GRU)



Placement	What changes	Matched control
S1 KAN head	$h' = h + \text{KAN}(\text{LN}(h))$ before dot-product scoring	+MLP head
S2 KAN input	$x' = x + \text{KAN}(\text{LN}(E[i]))$ before the GRU	+MLP input
S3 KAN-GRU cell	gates stay linear + sigmoid; candidate $n = \tanh(\text{KAN}([x, r \odot h]))$	MLP-GRU cell
Draft models	first-draft ungated Pure KAN and Hybrid, re-run fairly	Draft Hybrid (MLP)

KAN-NBR: for baskets, an additive KAN over 7 repeat-purchase features (recency, count, share, decayed frequency, popularity, overdue ratio, in-history) gives one plottable curve per feature.

4. Methodology: Steps and Deliverables

1 Data pipelines

YooChoose, Instacart, Dunnhumby with leakage checks

Deliverable: dataset tables, EDA

2 KAN module

4 bases, residual block, matched MLP, unit tests

Deliverable: kanrec/kan.py, 41 tests

3 Baselines

official GRU4Rec, basket-GRU, Pop, kNN, TopFreq, TIFU-KNN

Deliverable: reproduction check

4 KAN variants

head, input, cell; MLP controls; KAN-NBR

Deliverable: results per model and seed

5 Evaluation

Recall, MRR, NDCG, $P@|B|$, repeat/explore, Wilcoxon + Holm

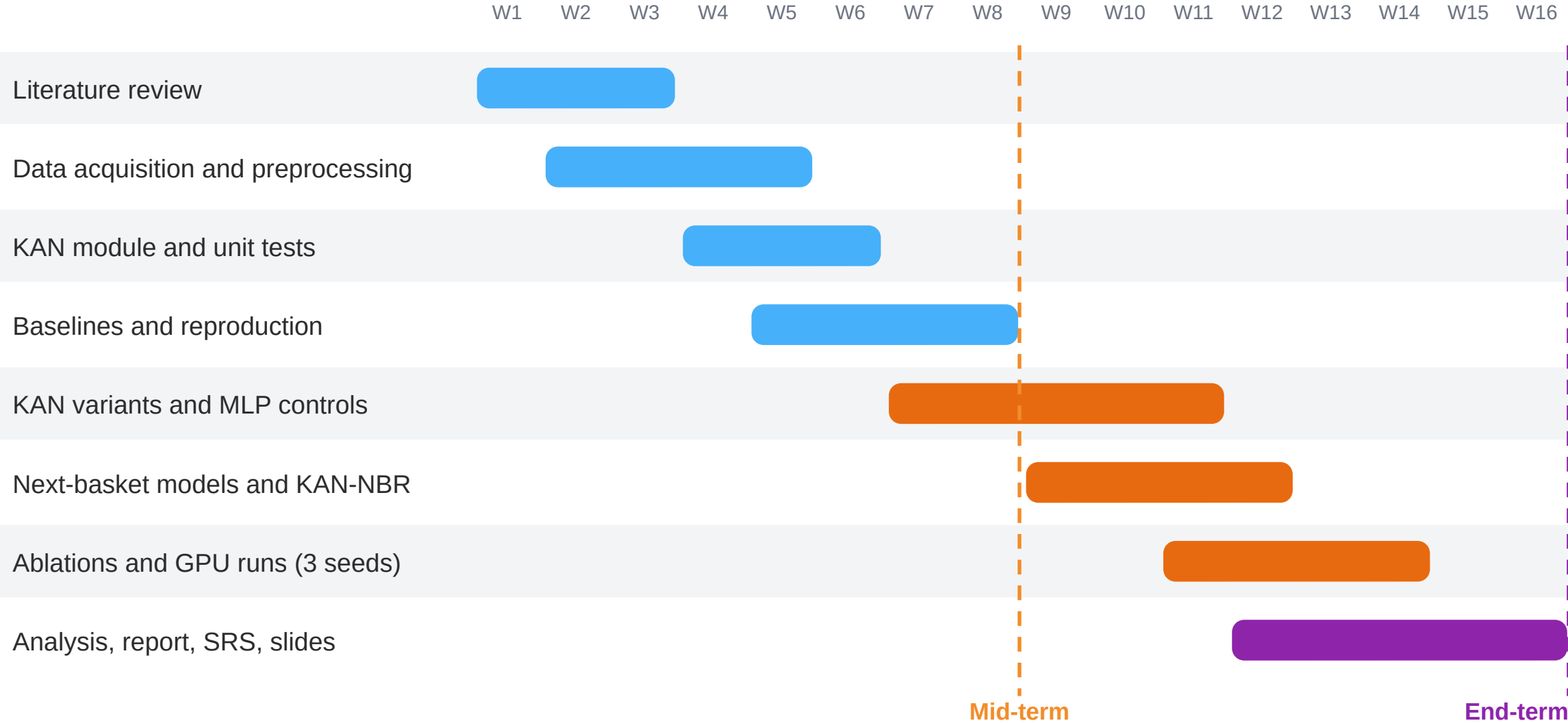
Deliverable: tables, figures

6 Report

notebook-generated tables in LaTeX, SRS, slides

Deliverable: report, SRS, decks

4. Methodology: Timeline



5. Working Model: Requirements and Architecture

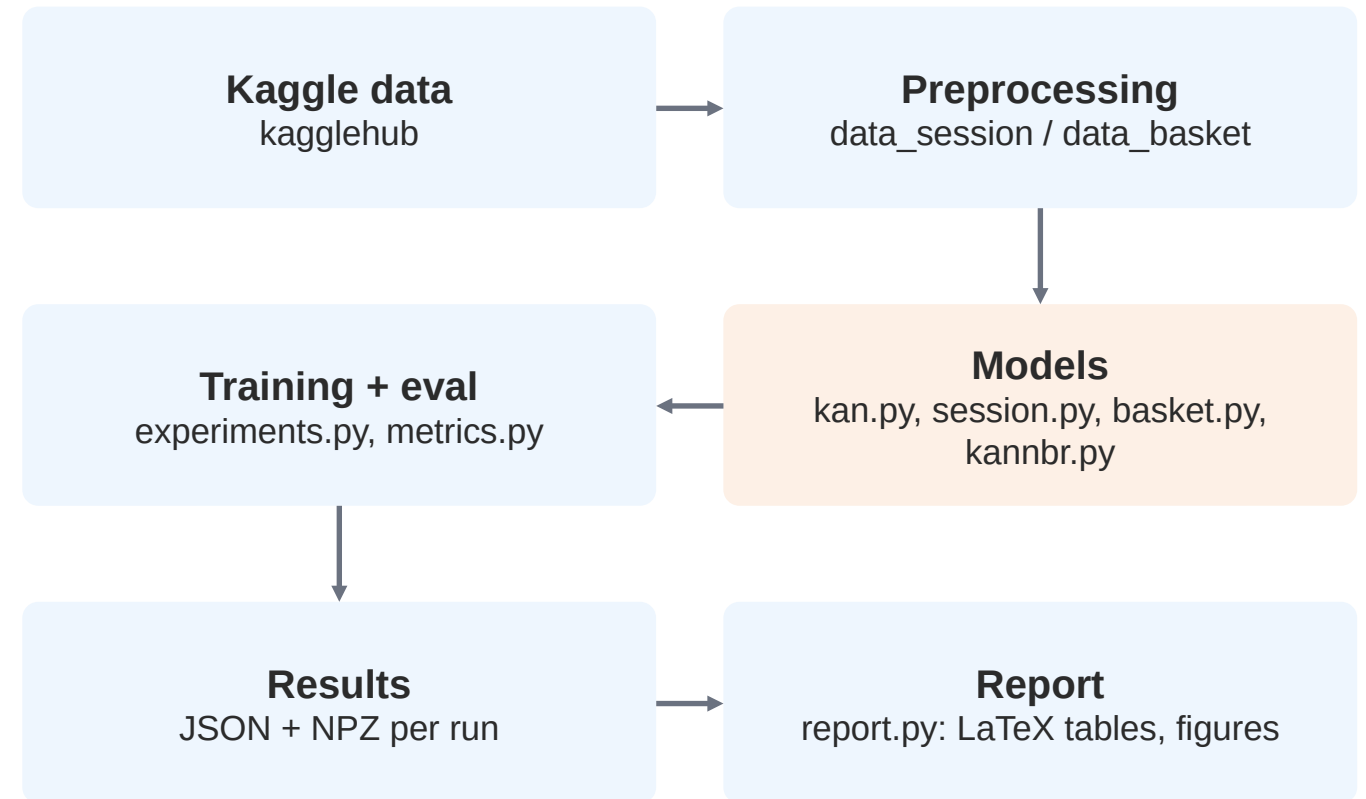
Requirement analysis

- Full SRS: KAN_Rec_SRS.docx
- Functional: download, preprocess, train, evaluate and report every model from one notebook
- Hardware profiles: Mac (Apple MPS), UPES Blackwell GPU, Google Colab
- Resumable: one result file per (dataset, model, seed)
- Software: Python 3.12, PyTorch, pandas, Jupyter, LaTeX

Code and results

github.com/yashdagar/kan-rec

Technical diagram: pipeline



5. Working Model: Attained Deliverables

41

unit tests passing

3

datasets with published protocols

0.690

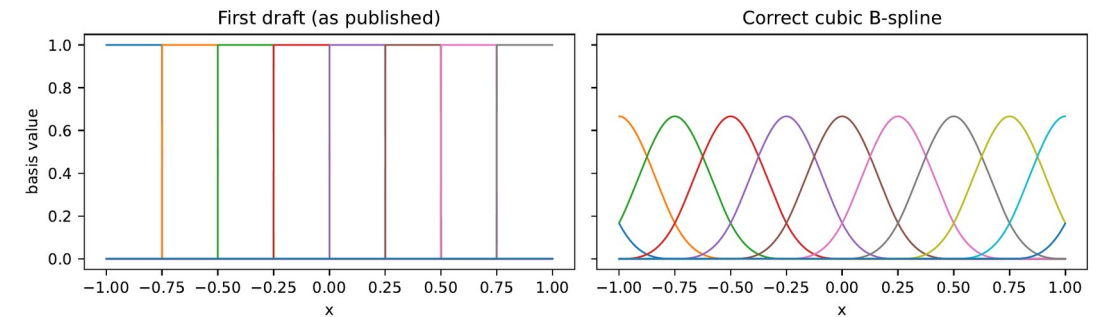
GRU4Rec Recall@20 (SR-GNN table: 0.606)

60

finished model runs (resumable)

Fixes to the first draft

- Leaky sample-level split replaced by session- and time-based splits
- Test set no longer used for model selection
- Degree-0 'B-spline' replaced by a verified cubic B-spline
- Ungated recurrent KAN replaced by gated KAN placements
- Every KAN paired with a parameter-matched MLP

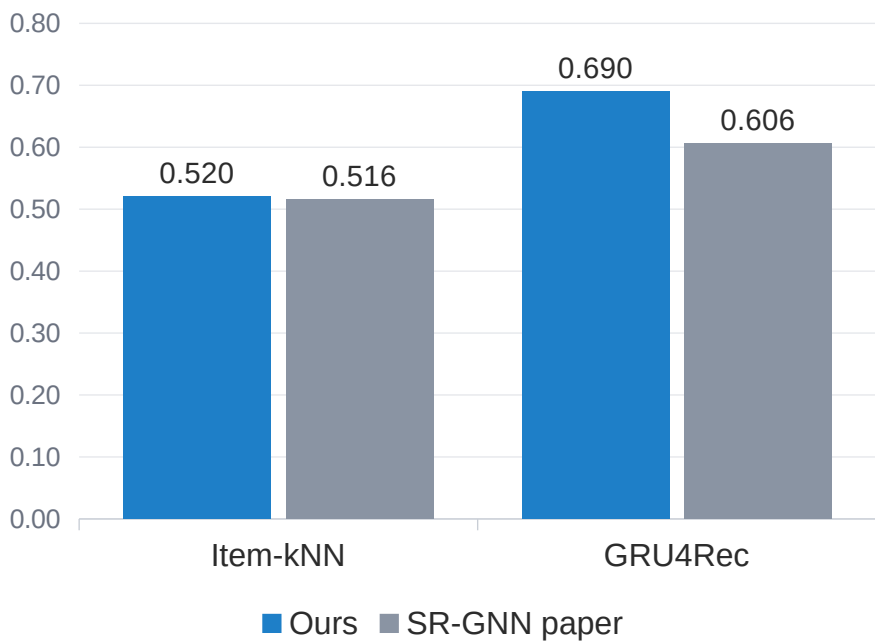


First-draft basis (step functions) vs. corrected cubic B-spline

6. Results: Test Cases and Reproduction

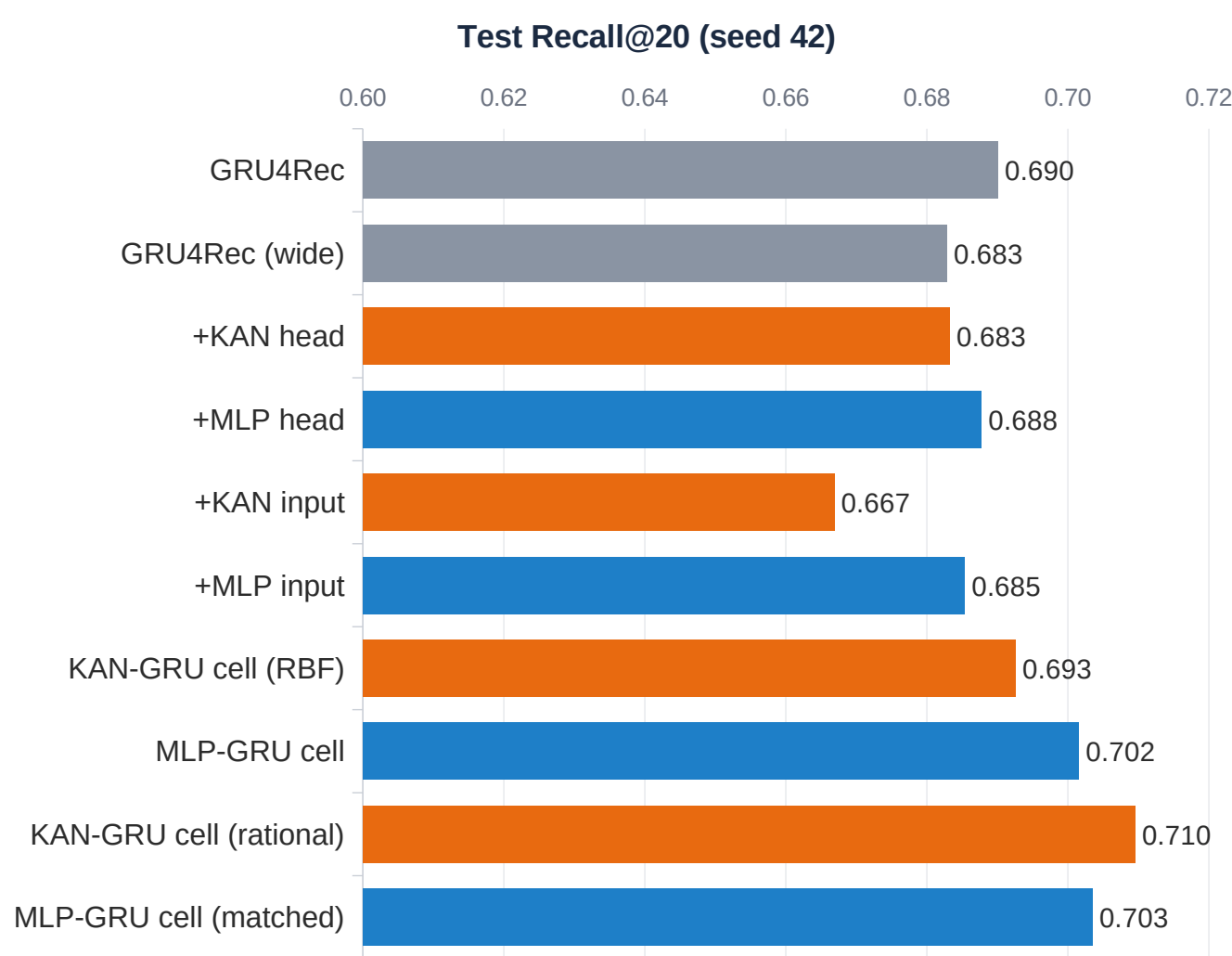
Test case (pytest)	Checks	Status
B-spline basis	partition of unity; equals closed-form cubic	Pass
KAN layer	shapes, gradients, parameter count formula	Pass
Matched MLP	parameters within 2% of the KAN it controls	Pass
GRU4Rec wrapper	identical outputs to the official model	Pass
Session iterator	same batches as the official iterator	Pass
Metrics	toy rankings; vectorised = loop version	Pass
Splits	no session or user in two splits; time order	Pass
KAN-NBR	batched scores = per-user scores	Pass

Reproduction (YooChoose 1/64)



Test Recall@20. Our GRU4Rec uses the tuned official settings (Hidasi & Karatzoglou 2018), so it exceeds the older number in the SR-GNN table.

6. Results: Session-Based (YooChoose 1/64)



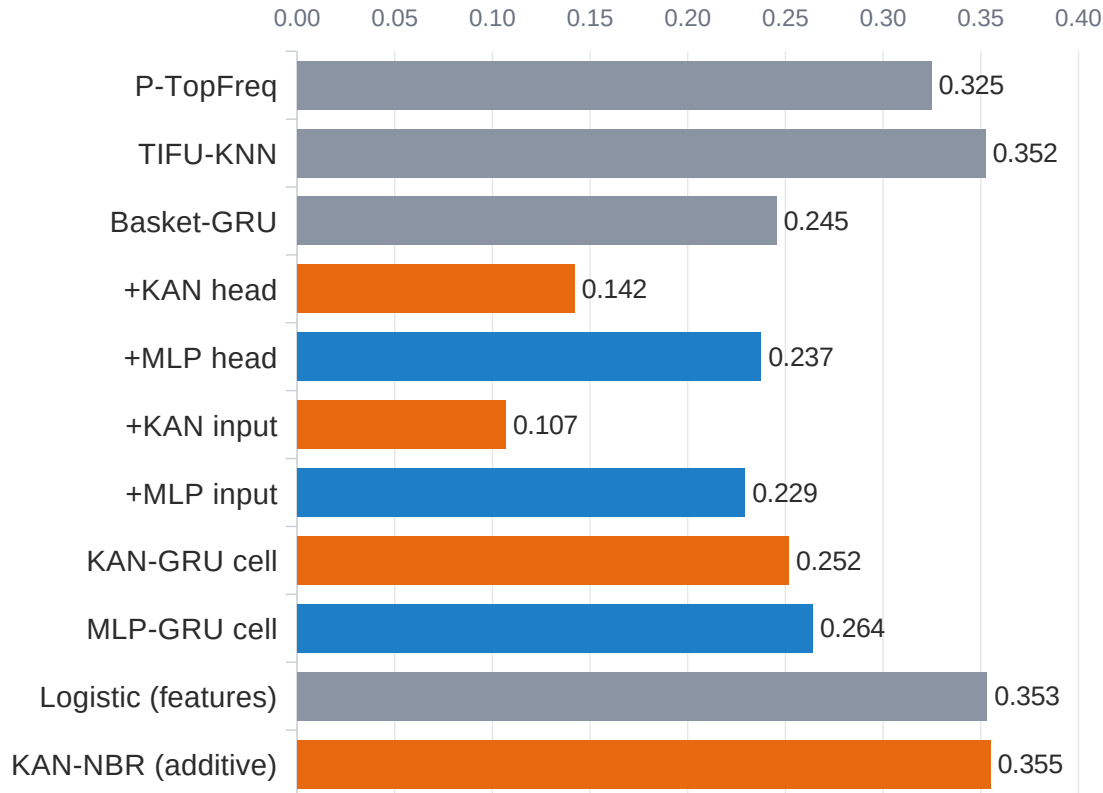
Baseline KAN Matched MLP

Findings

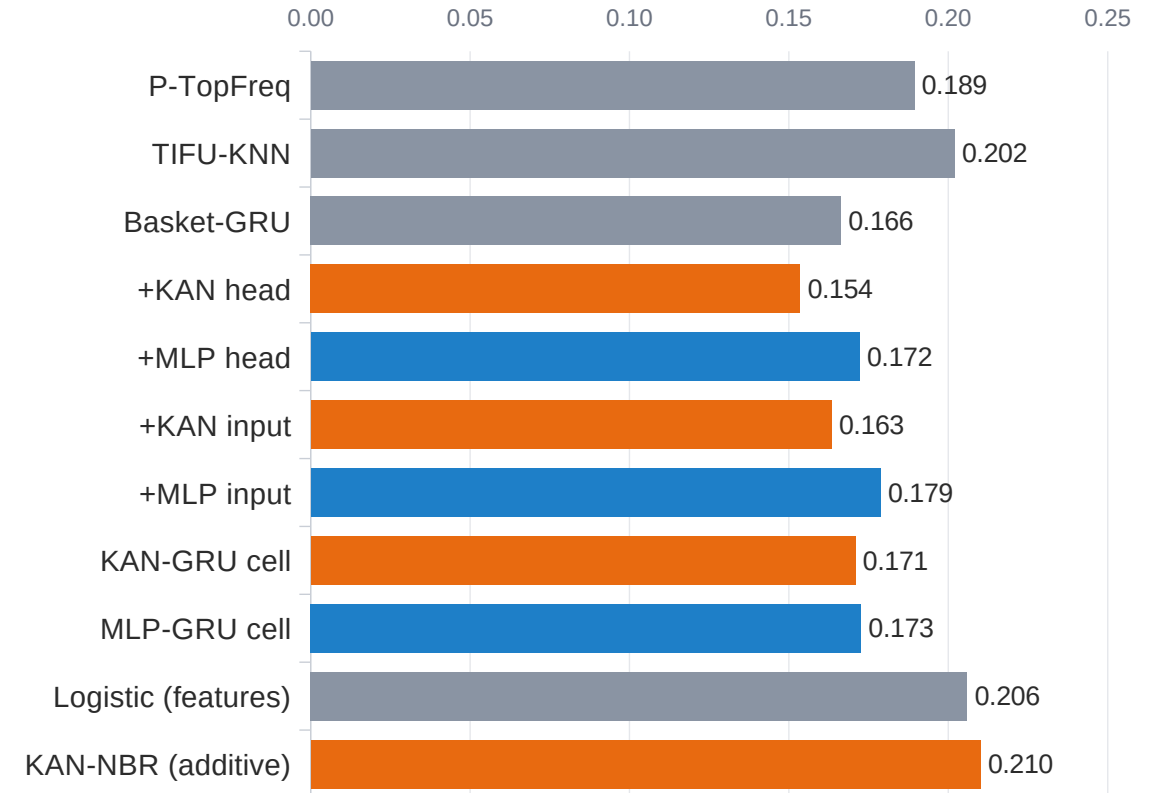
- KAN head and KAN input do not beat their MLP controls
- Putting the non-linearity inside the GRU cell helps both KAN and MLP
- Best: KAN-GRU cell with rational basis, **0.7095** vs matched MLP 0.7035 and GRU4Rec 0.6901
- A wider GRU4Rec with the same parameters does not help, so the gain is not just size
- Seed 43 agrees; third seed pending

6. Results: Next-Basket (Instacart, Dunnhumby)

Instacart: test Recall@10

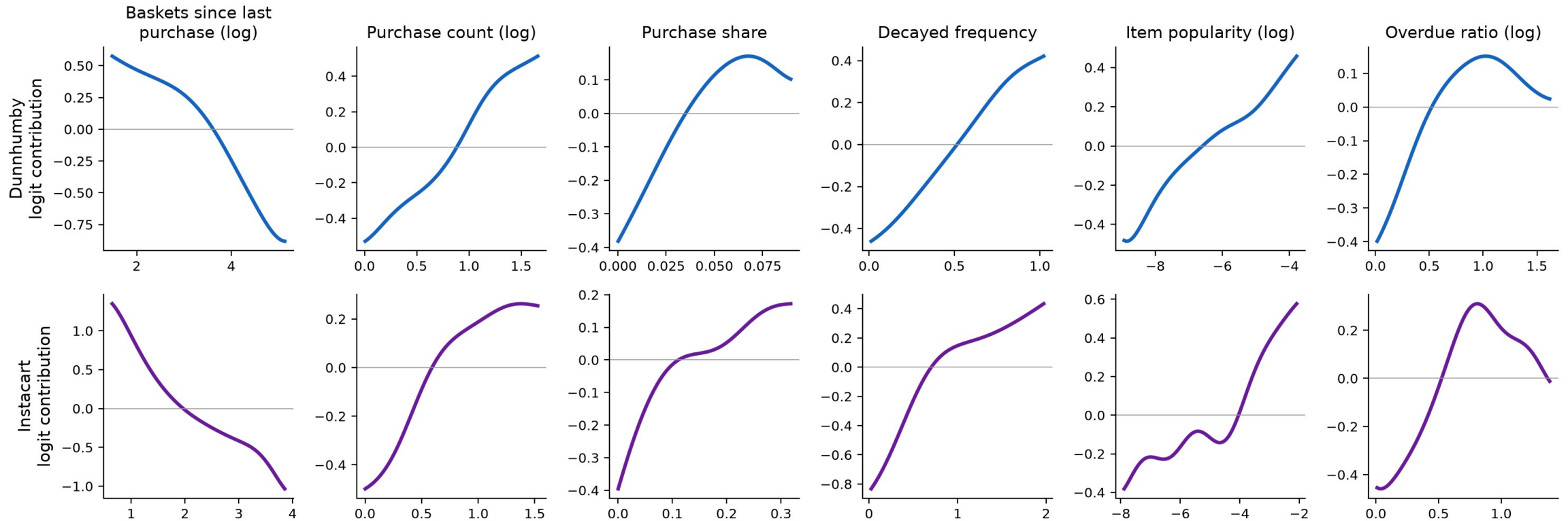


Dunnhumby: test Recall@10



Finding: KAN head and input blocks hurt the basket-GRU; a KAN-GRU cell helps a little but its matched MLP helps as much or more. Frequency models win, and KAN-NBR matches the best of them while staying interpretable.

6. Results: Interpretability of KAN-NBR



Each curve is one learned KAN function: its contribution to the purchase logit.

- **Overdue ratio:** the purchase probability peaks when an item is about one usual interval overdue, then falls; a logistic model can only fit a straight line here.
- **Recency and frequency:** recent and frequent items score higher, with diminishing returns.

6. Results: KAN vs Matched MLP

Dataset	Placement	Metric	KAN	MLP	Winner
YooChoose	Head	Recall@20	0.6832	0.6878	MLP
YooChoose	Input	Recall@20	0.6669	0.6854	MLP
YooChoose	GRU cell (RBF)	Recall@20	0.6926	0.7016	MLP
YooChoose	GRU cell (rational)	Recall@20	0.7095	0.7035	KAN
Instacart	Head	Recall@10	0.1420	0.2375	MLP
Instacart	Input	Recall@10	0.1067	0.2293	MLP
Instacart	GRU cell	Recall@10	0.2517	0.2639	MLP
Dunnhumby	Head	Recall@10	0.1535	0.1722	MLP
Dunnhumby	Input	Recall@10	0.1634	0.1787	MLP
Dunnhumby	GRU cell	Recall@10	0.1709	0.1726	tie
Dunnhumby	KAN-NBR vs MLP-NBR	Recall@10	0.2101	0.2050	KAN

2 KAN better

8 MLP better

1 within 0.002

Same parameter count in every pair; single seed (Mac profile), difference threshold 0.002.

7. Conclusion: Justification of Objectives

Objective	Outcome
Fix first-draft errors	Done: new splits, verified B-spline, gated placements, errata appendix
KAN layer, four bases	Done: unit-tested; rational basis is the most efficient
KAN in GRU4Rec	Head/input: no gain over MLP. Cell (rational): best Recall@20 on YooChoose
KAN in basket-GRU	No gain; frequency methods dominate next-basket accuracy
Fair comparison	Matched MLPs, wide GRU, significance tests, published protocols
KAN-NBR	Matches TIFU-KNN-level accuracy with readable feature curves

Take-away: KAN layers are not a free accuracy gain. Where they help, an equally large MLP usually helps as much; their clearest value is interpretability on small, meaningful inputs.

7. Conclusion: Future Scope

1 GPU runs

three seeds, yoochoose1/4 and full Instacart on the UPES Blackwell GPU

2 Attention models

group-rational KAN in the feed-forward block of SASRec

3 Covariates

KAN curves over time of day and days since last order

4 More data

Diginetica as a second session benchmark

5 KAN training

grid extension during training; faster fused kernels

6 Deployment

a small KAN-NBR service for interpretable basket suggestions

A: Paper published. None yet; a paper on the KAN-GRU cell and KAN-NBR results is in preparation.

B: Cited papers

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Thank You